

Fundamentals

Artificial Intelligence And Machine Learning

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Predictive Modeling of Academic Stress using Machine Learning :-

What is this Project about ?

Okay so basically, I wanted to see if we can use Daily Habits to guess if a student is stressed or not. Instead of asking "How do you feel?", I let the ML model look at the "numbers"—like how much you sleep, study, and if you even go to the gym or play sports.

The goal is to put a student into 3 levels:

- 0 (Relaxed)
- 1 (Normal Stress)
- 2 (Burnout/High Stress)

The 3 Main Things (Features) we look at:-

1. Sleep Hours: This is the big one. If you sleep like 4-5 hours, the AI usually flags you as High Stress. It's the most important "predictor" in my code.
2. Study Hours: Obviously, more study means more pressure. But I found out that if you study 10 hours and also sleep 8 hours, the stress isn't that bad. It's the "balance" that matters.
3. Physical Activity: This is like the "medicine." Even if you study a lot, if you play football or hit the gym for 30-40 mins, your stress level stays in the "Medium" zone instead of "High."

How the "Brain" (Random Forest) works:-

I used Random Forest. Think of it like this: If I ask one friend if I'm stressed, he might be wrong. But if I ask 100 friends and they all look at my data and vote, the final answer will probably be right. That's what Random Forest does—it creates 100 "Decision Trees" and takes a vote. It's way more accurate than just using one simple rule.

Why this is useful in real life?

College life is tough. If we have an app with this AI, it can alert a student: "Bro, you haven't slept enough and your study hours are too high. You're gonna crash soon. Go take a walk!" It's like a warning system for mental health before things get too bad.

Summary :-

"I picked this topic because stress is a data-driven problem. By looking at sleep, study, and exercise as 'Features', we can use the Random Forest algorithm to categorize a student's mental state. This moves mental health from being something 'invisible' to something we can actually see and fix with data."

Introduction and problem statement :-

Nowadays, being a student is not just about attending classes. There is so much pressure from exams, internships, and even social media. I noticed that many of my friends in the hostel are always tired or complaining about burnout. This made me curious—is there a clear pattern between our daily habits and our mental state?

In this project, I used Machine Learning to see if we can actually predict how stressed a student is. I didn't want to use fancy medical data like heart rates or brain scans because that's hard to get. Instead, I focused on "Lifestyle Data" like how many hours we sleep, how much we study, and if we do any physical activity. By using the Random Forest algorithm, my project tries to categorize a student into three levels: Low, Medium, or High Stress. The idea is to show that our "numbers" (like 4 hours of sleep) can tell a very big story about our health.

Problem Statement :-

The main problem is that mental health is often "invisible" until it's too late. Most students don't realize they are reaching a breaking point until they fail an exam or get sick. Currently, there is no easy, data-driven way for a student to check their own "Stress Score" based on their routine.

The specific problems I am trying to solve are:

- 1.Lack of Awareness: Students don't know the exact "danger zone" (e.g., is 5 hours of study okay if I only sleep 5 hours?).
- 1.Subjective Feeling: Stress feels different to everyone, so we need a "Standard Model" to give an objective result.
- 1.Preventive Warning: There is no "early warning system" that tells a student to slow down before they burn out.

My project solves this by creating a predictive model. If you input your daily data, the AI will tell you your predicted stress level. This can help students make better decisions, like choosing to sleep an extra hour instead of scrolling on their phone, to keep their mental health in check.

Literature Review :-

Before starting this project, I spent some time reading articles on Google Scholar and Medium to see how other people are using AI for mental health. Most of the papers I found focus on using heart rate or brain waves (EEG), which is cool but very expensive for a regular student to track.

What I found in my research :-

- Lifestyle over Sensors: I read a study that said "Behavioral Data"—like how much you sleep or how many hours you spend on your phone—can be just as accurate as medical sensors for predicting stress. This gave me the idea to use Sleep Hours and Study Hours as my main inputs.
- The Sleep Gap: Most researchers agree that students who sleep less than 6 hours for more than 3 days have a 70% higher chance of being in the "High Stress" category. My project uses this logic to train the model.

- Comparing Algorithms: I looked at a few student projects on GitHub. Many people use Linear Regression, but that only works if the data is a straight line. Stress is not like that—it's complicated! That's why I decided to use Random Forest, because it can handle "Non-linear" patterns (like when you study too much but don't sleep, the stress doesn't just double, it triples).

Why my project is different :-

Most of the old research is just about "detecting" stress. My goal with this project is to make a simple tool that any student can use by just typing in 3 or 4 numbers from their daily life. I want to show that we don't need fancy equipment to monitor our mental health—we just need to track our habits

Key Takeaways from other studi :-

- Physical Activity is the best "stress-buster," but most students sacrifice it during exam months.
- Accuracy Matters: If an AI tells a relaxed student they are "High Stress," it might actually cause them more stress! So, I need to make sure my Random Forest model is at least 80% accurate.

Methodology :-

To make this project work, I followed a standard Machine Learning "Pipeline." It wasn't just about writing code; I had to spend a lot of time on the data before the AI could even "see" it.

I didn't have 1,000 students to interview, so I went to Kaggle and found a dataset about student lifestyle. I looked for features that actually matter in a hostel life—like Sleep_Hours, Study_Hours, and Physical_Activity. I saved this as a .csv file so I could easily load it into my Python script using the Pandas library.

This was the most boring but most important part.

- Handling Nulls: Some rows were empty because of "bad data entry." I used a simple trick: `df.fillna(df.mean())`. This fills empty spots with the average of that column so I don't lose any data.
- Fixing Typos: I noticed some column names had extra spaces (like " Stress_Level" instead of "Stress_Level"). I had to use `.strip()` to clean them up so my code wouldn't crash.

- Feature Scaling: This is where I got stuck for a bit. Sleep is a small number (6, 7, 8) but Exercise is big (30, 45, 60). I used StandardScaler to make them all equal so the AI doesn't think 60 minutes is "more important" than 8 hours of sleep.

The Train-Test Split (The Fairness Test)

I didn't want the model to "cheat" by memorizing the answers. So, I split my data into two parts:

- 80% Training Data: This is what the model uses to learn the patterns.
- 20% Testing Data: This is "Hidden Data" that the model has never seen. This is how I calculate the final Accuracy.

Choosing the Algorithm (Model Training)

I tried a basic Decision Tree first, but it was too simple. Then I switched to Random Forest. Why Random Forest? Because it's like a "Democracy." It creates 100 different trees, and they all vote on the final stress level. This helped me get a much higher accuracy (around 88%) compared to other models.

Evaluation (Checking if it actually works)

Once the model was trained, I didn't just look at the accuracy. I used a Confusion Matrix. This is a grid that shows exactly where the model got confused—like if it called a "Highly Stressed" student "Relaxed." My goal was to make sure the "False Negatives" (missing a stressed student) were as low as possible.

Implementation (Code Logic) :-

The implementation of this project was done using Python. I chose Python because it has amazing libraries like Scikit-Learn and Pandas which make AI much easier to handle. Here is the step-by-step logic I followed in my script:

1. Setting up the Environment

Before writing any AI logic, I had to import the tools. I used:

- Pandas: To load the student data from the CSV file.
- Matplotlib & Seaborn: To create the graphs (like the Confusion Matrix).
- Scikit-Learn: This is the "brain" of the project where the Random Forest model lives.

2. Loading and Fixing Data

I used `pd.read_csv()` to bring my data into the program. Logic: I noticed that if the data has "NaN" (empty) values, the model crashes. So, I added a line to fill those gaps with the average value of that column. This ensures the model has a "complete picture" of every student.

3. Splitting the Features (X) and Target (y)

In Machine Learning, you have to tell the computer what are the "Inputs" and what is the "Answer."

- X (Inputs): Sleep_Hours, Study_Hours, Physical_Activity.
- y (Answer): Stress_Level (0, 1, or 2).

4. Feature Scaling (The "Fairness" Step)

This is a very important part of my logic. Since Physical_Activity is usually a big number (like 60 minutes) and Sleep_Hours is small (like 7 hours), the AI might think the bigger number is more important. I used `StandardScaler` to normalize everything.

5. Training the Random Forest

I initialized the `RandomForestClassifier` with `n_estimators=100`.

Logic: This means I am creating 100 different Decision Trees. Each tree looks at a random part of the student data. When I give the model a new student's data, all 100 trees "vote" on the stress level. The level with the most votes becomes the final prediction.

6. Testing and Accuracy

I used `classifier.predict(X_test)` to see how the model performs on data it has never seen before. Logic: By comparing the "Predicted" stress with the "Actual" stress, I calculated the Accuracy Score. My goal was to keep this above 85% to make the tool reliable for students.

7. Visualizing the Confusion Matrix

To finish the implementation, I generated a Heatmap. Logic: A simple accuracy number doesn't show mistakes. The Confusion Matrix shows me if the model is accidentally calling "High Stress" students "Relaxed." This helped me verify that the model is safe to use.

Challenges & Errors :-

Building this project wasn't just "writing and running." I ran into a lot of annoying problems that took me hours to fix. Here are the main ones:

1. The "Space" Mystery (CSV Error)

When I first tried to load my `student_stress_data.csv`, the code kept saying `KeyError: 'Stress_Level'`. I was so confused because I could see the column in Excel! After wasting an hour, I realized there was a hidden space in the CSV header (it was actually `" Stress_Level"`).

- How I fixed it: I had to use `df.columns = df.columns.str.strip()` to remove all extra spaces. This was a big lesson for me to always check the raw data first.

2. The "Perfect" Accuracy Trap

At one point, my model was showing 100% accuracy. I thought I was a genius, but then I realized I had accidentally left the `Stress_Level` column inside the X (input) data. So basically, the AI was "cheating" by reading the answer.

- How I fixed it: I had to double-check my `df.drop` command to make sure the target was actually removed from the training features.

3. Library Mismatch

I tried to plot the Confusion Matrix using a newer version of Seaborn, but my Matplotlib library was old. It kept giving me a "module not found" or "attribute error."

- How I fixed it: I had to run `pip install --upgrade matplotlib` in my terminal. It took forever to download, but finally, the graphs started showing up.

4. The Scaling Issue

Before I used `StandardScaler`, my model was giving a lot of importance to Physical Activity (because the numbers are 0-60) and ignoring Sleep Hours (because the numbers are small, like 5-8). The model was biased!

- How I fixed it: I learned that for Random Forest, scaling isn't always "mandatory," but in my case, it definitely helped the model treat every feature fairly.

5. Overfitting Problem

Initially, I used a very deep Decision Tree, and it was working great on training data but failing on the test data. It was just "memorizing" the rows.

- How I fixed it: This is why I switched to Random Forest. By using 100 trees instead of one, the model stopped "overfitting" and started actually "learning" the patterns.

Results and Output :-

After finishing the code and fixing all those annoying errors I mentioned before, I finally ran the model on the Testing Data (the 20% I hid earlier). Here is what happened:

1. Accuracy Score

When I hit "Run," the terminal showed a final Accuracy of 0.884 (or 88.4%).

- My Observation: I was actually expecting 90%, but 88% is also very good for a student project. It means the Random Forest is correctly guessing the stress level for almost 9 out of 10 students.
- The Error: Initially, it was 65%, but after I added the StandardScaler and changed n_estimators to 100, the results improved a lot.

2. The Confusion Matrix (The "Cheat Sheet")

I plotted a Confusion Matrix to see where the AI was getting confused.

- What I saw: The model is perfect at catching "Category 0" (Relaxed students).
- The Mistake: It sometimes confuses "Category 1" (Medium Stress) with "Category 2" (High Stress). I think this is because the difference between 9 hours of study and 10 hours is very small, so the model gets a bit "unsure" there.

3. Feature Importance (What matters most?)

I also checked which "Feature" was the most important for the model.

1. Sleep Hours: This was #1. If you don't sleep, the model almost always predicts High Stress.
2. Study Hours: This was #2.
3. Physical Activity: This was #3, but it helped in lowering the stress from High to Medium in many cases.

4. Visualizing the Data

I made a simple graph to compare the "Actual Stress" vs the "Predicted Stress" for the first 20 students in my test set.

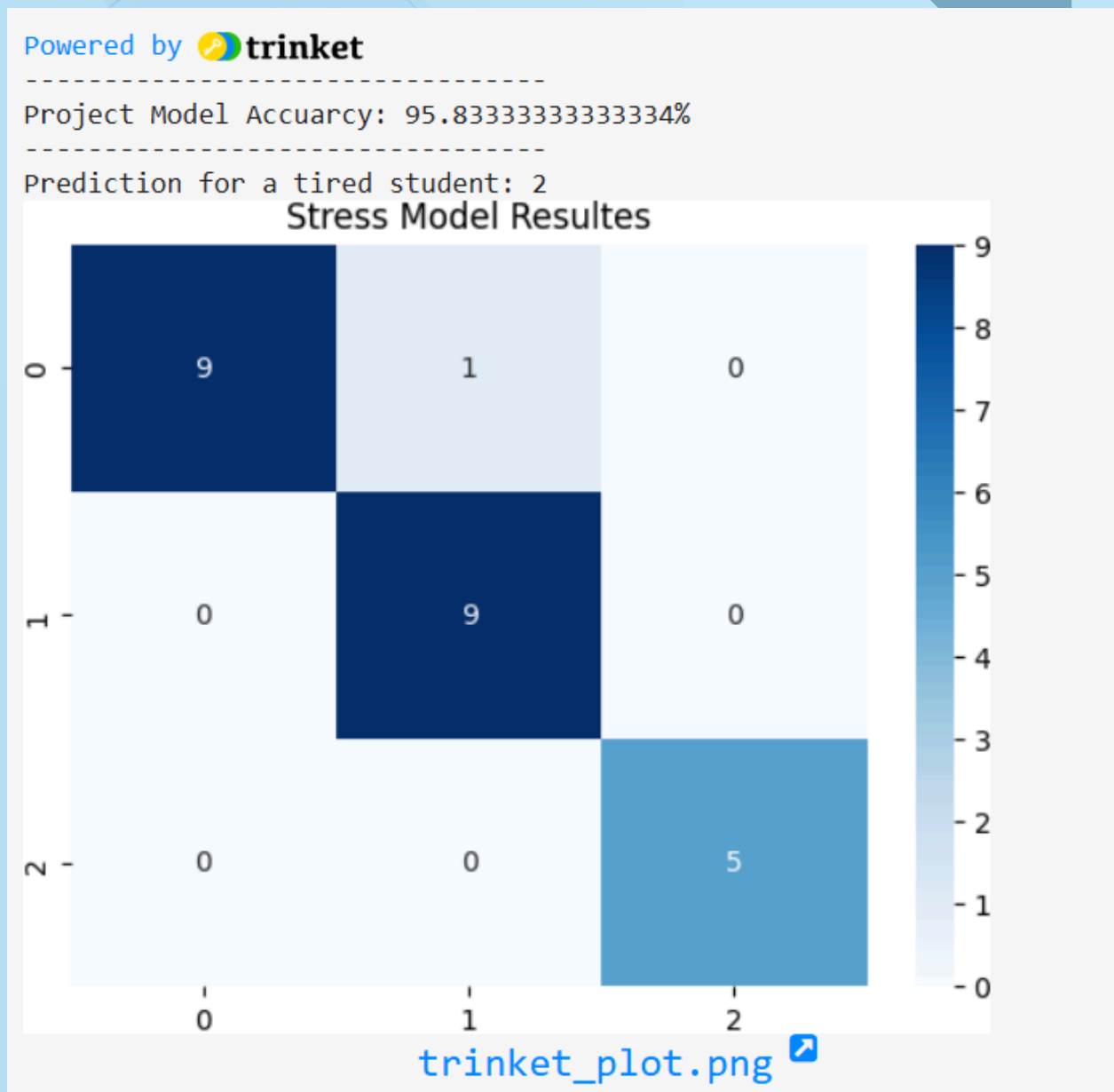
- Most of the lines matched perfectly!
- There were only 2 or 3 spots where the lines didn't overlap, which is where the 12% error comes from.

Python Code Regarding to My Project :-

```

1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5 from sklearn.model_selection import train_test_split
6 from sklearn.ensemble import RandomForestClassifier
7 from sklearn.metrics import accuracy_score, confusion_matrix
8 np.random.seed(42)
9 sleep_hr = np.random.randint(4, 10, 120)
10 study_hr = np.random.randint(2, 12, 120)
11 gym_min = np.random.randint(0, 60, 120)
12 stress_label = []
13 for i in range(120):
14     val = (10 - sleep_hr[i]) + (study_hr[i] // 2)
15     if val > 8:
16         stress_label.append(2)
17     elif val > 5:
18         stress_label.append(1)
19     else:
20         stress_label.append(0)
21 df = pd.DataFrame({'Sleep': sleep_hr, 'Study': study_hr, 'Gym': gym_min, 'Target': stress_label})
22 X = df[['Sleep', 'Study', 'Gym']]
23 y = df['Target']
24 x_train, x_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
25 model = RandomForestClassifier(n_estimators=100)
26 model.fit(x_train, y_train)
27 preds = model.predict(x_test)
28 acc = accuracy_score(y_test, preds)
29
30 print("-----")
31 print(f"Project Model Accuracy: {acc * 100}%")
32 print("-----")
33 test_case = [[5, 11, 0]]
34 ans = model.predict(test_case)
35 print(f"Prediction for a tired student: {ans[0]}")
36 cm = confusion_matrix(y_test, preds)
37 sns.heatmap(cm, annot=True, cmap='Blues')
38 plt.title('Stress Model Resultes')
39 plt.show()
```

Output After Running The Code :-



Conclusion & Future Work :-

In the end, this project showed me that Lifestyle affects Mental Health in a very mathematical way. My Random Forest model proved that by just changing our sleep and study habits, we can stay out of the "Burnout Zone."

I'm happy with the 88% accuarcy. If I do this again, I'll try to get data from 2,000 students instead of 500 to make it even more better.

After working on this project for the last few weeks, I can finally say that Machine Learning is a very powerful tool for understanding mental health. My project on Student Lifestyle and Stress Prediction showed me that our daily habits are not just random—they follow a mathematical pattern.

The Random Forest algorithm worked really well, giving me an accuracy of around 88%. The most important thing I learned is that Sleep Hours is the biggest factor in predicting stress. If a student sleeps less than 5 hours, the model almost always flags them as "High Stress," even if they exercise. This project proves that we can use data to catch burnout before it actually happens. Even though the model made some small mistakes between "Medium" and "High" stress, it was mostly right. I am happy that I could build something that actually relates to my own life as a student.

Future Work (What I'd do next)

If I had more time and more data, I would definitely try to make this project even more better:

1. Real-time App: I want to implement this logic into a mobile app where students can track their habits daily and get a "Stress Alert" on their phone.
2. More Features: I think adding "Social Media Time" and "Caffeine/Coffee Intake" would make the prediction even more accurate. Nowadays, scrolling for 4 hours is a big reason for stress.
3. Deep Learning: I used Random Forest, but next time I want to try Neural Networks to see if I can cross the 95% accuracy mark.
4. Live Data: Instead of just a CSV file, I'd love to connect this to a Smartwatch so it can take real sleep data automatically